

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Eighth Semester of B. Tech (CL) Examination****April - May 2018****CL 414 Construction Technology****Date: 02.05.2018, Wednesday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I**Q.1 Do as directed. [07]**

- i. Describe heaped capacity and struck capacity of bucket with illustration.
- ii. Screw Take-ups in Belt Conveyor system allows to stretch and shrink the length of the belt. True or false?
- iii. Justify the statement "If any machine will be operated at high altitude, it will cause a decrease in available engine power".
- iv. As a site engineer, which equipments would you employ for excavation for highway construction project in rocky hilly terrain?
- v. Which equipment is capable to dig and haul the material simultaneously, at certain distance?
- vi. _____ equipment is used for digging trench for basement construction of multistory building.
- vii. What are the various operations involved in digging with power shovel?

Q.2 (a) What do you mean by rolling resistance on haul road? Enlist and explain factor affecting the rolling resistance. [03]**(b) The soil borrow material to be used to construct a highway embankment has a mass unit weight of 1525 kg/m^3 , a water content of 9%, and the specific gravity of solids is 2.64. The specifications require that the soil be compacted to dry unit weight of 1634 kg/m^3 and the water content be held to 12%. [07]**

- i. How many cubic meter of borrow material is required to construct an embankment having 150000 m^3 net section volume?
- ii. How many liters of water is required to added to the borrow material, on a per m^3 basis? Assuming no loss by evaporation.

OR**(b) Do as directed.**

- i. The haul road from borrow pit to the fill has an adverse grade of 4 %. Wheel type hauling [03]

unit will be used on job. It is expected that haul road rolling resistance will be 50 kg/ton. What will be effective grade for haul? Will the unit experience the same for the return trip?

- ii. A tractor weighs 12.4 ton & has maximum rim pull in first gear is 6762 kg. If it is operated on a haul road with positive slope of 2 % & rolling resistance of 50 kg/ton, what will be pull available for the towing load? [04]
- (c) Write a note on different types of bulldozer blade and attachments of bulldozer blade. [04]

OR

- (c) Discuss the factors affecting the output of the dragline. [04]
- Q.3 (a)** Classify and describe in short construction equipments on a functional basis with sub grouping for construction projects. [04]
- (b) Attempt any two.** [10]
- i. From the following information, determine the cost of production (excavation & hauling) in terms of rupees per cubic meter.
- **Excavating equipment:** Hoe with 1.33 m³ dipper (can handle 1.51 m³) having cycle time of 16 seconds and operating factor 55 minutes per hour, cost Rs. 3500/hour
 - **Material:** Good common earth with swell of 20% and fill factor 0.85
 - **Hauling units:** Trucks 8.5 m³ capacity with operating factor of 50 minutes per hour and having round trip time 22 minutes, cost Rs. 400/hour.
- ii. In which situations the belt conveyors are used? Explain function of following components of belt conveyor.
- Idlers
 - Snub pulley
 - Drive pulley
- iii. Discuss the role of tractor in earth moving. What considerations have governed the selection of wheel type or crawler type tractor on a job? Compare the application of two types of tractor.

SECTION - II

- Q.4 Do as directed.** [07]
- i. Enlist the reason for replacement of an equipment.
- ii. Cofferdams are permanent enclosures to keep out water and soil so as to permit dewatering and construction of the permanent structure in dry environment. True or False?

- iii. Which cost would you consider to workout operating cost of an equipment?
- iv. Define “Depreciation”.
- v. Enlist types of shoring system provided to support unsafe structure.
- vi. The operation of removing the formwork is known as _____.
- vii. In double scaffolding, the first row of standards is _____cm. away from the wall & second row of standards is _____ m. away from the first row.

- Q.5 (a)** Differentiate between shotcrete and conventional concrete. What are the advantages of shotcrete? **[04]**
- (b)** Write a short note on Tremie method of underwater concreting. **[05]**

OR

- (b)** Enlist and explain different types of Ready mixed concrete. What are the features of RMC? **[05]**
- (c)** Explain the sequence of construction of coffer dam. **[05]**

OR

- (c)** Enlist types of geo-textiles. What are the applications of geo-textiles? **[05]**

- Q.6 (a)** Enlist various methods of soil stabilization and explain any one in detail. **[04]**
- (b)** Enlist different types of construction projects where Slipform technology can be used. What are the advantages of Slipform technology? **[05]**

OR

- (b)** Enlist types of scaffolding. Explain single scaffolding in detail with sketch. **[05]**
- (c)** An investing firm is intending to give equipment on rent to contractors. If MARR is 15%, determine the probable hourly rental cost of diesel engine power shovel from the following data. **[05]**

Factory Price	: Rs. 9,50,000
Useful Life	: 8 Years
Salvage value	: 12 % of original cost
Freight charges	: Rs. 20/ton
Engine	: 140 Hp
Shipping weight	: 120 ton
Crank case capacity	: 20 liters

Operating factor	: 0.75
Unloading and assembly charges:	: Rs. 5000
Time between oil changes	: 120 hours
Maintenance and repair cost	: 70 % of annual depreciation
Hours used per year	: 6000 hours

OR

- (c) As an owner of equipment, how would you decide economic life of equipment? Discuss [05]
the factors to be considered while deciding the economic life of equipment.
